PHARMACEUTICAL CASE STUDY

Tablet Formulation & Process Engineering

An end-to-end technical analysis of oral solid dosage form design, excipient matrix functional selection, stability kinetics, and scale-up manufacturing challenges.

Dosage Form: Immediate-Release Film-Coated Tablet

Model API: Low-Solubility / High-Dose Active

Dosage Selection & Target Profile

Why Oral Compressed Tablets?

Oral tablets account for over 60% of all prescribed pharmaceuticals due to unmatched patient compliance, accurate dosing precision, shelf-life stability, and cost-effective mass production.

Target Product Profile (TPP)

- **Dose Strength:** 500 mg active ingredient per unit.
- **Release Profile:** Immediate Release (>85% in 30 min).
- **Friability:** Below 0.8% weight loss after 100 rotations.
- **Shelf Life Goal:** 24 to 36 months under Zone IV climate.

Excipient Matrix & Functional Roles

Functional Class	Selected Excipient	Specific Role & Mechanism	Concentration
Diluent / Filler	Microcrystalline Cellulose (MCC PH-102)	Provides bulk volume, excellent compressibility, and plastic deformation during compaction.	30 – 45% w/w
Binder	Povidone (PVP K-30)	Promotes interparticle cohesion during wet granulation to ensure granule mechanical strength.	2 – 5% w/w
Disintegrant	Croscarmellose Sodium (Ac-Di-Sol)	Facilitates rapid tablet breakup via high cross-linked swelling and water wicking action.	3 – 6% w/w
Glidant & Lubricant	Colloidal Silicon Dioxide & Mg Stearate	Reduces powder friction, enhances blend flowability, and prevents punch face sticking.	0.5 – 1.5% w/w
Film Coating	HPMC / PEG (Opadry II System)	Provides moisture barrier protection, taste masking, and smooth aesthetic swallowing surface.	2 – 3% weight gain

End-to-End Manufacturing Process

1. Dispensing & High-Shear Granulation
API and excipients are dry-mixed, followed by binder fluid addition to form cohesive wet granules.

2. Fluid Bed Drying & Cone Milling
Wet mass is dried to target Loss on Drying (LOD < 2.0%) and milled for uniform particle size.

3. Compression & Pan Coating
Lubricated blend is compressed on high-speed rotary press and film-coated in perforated pan.

Stability Factors & Degradation

Chemical Pathways

Preventing hydrolysis and oxidation through protective film coating and low residual moisture.

$$+ H_2O \rightarrow$$

Physical Stability

Monitoring tablet hardness, friability changes, and polymorph transitions over shelf life.

$$\text{Friability \%} = \frac{W_1 - W_2}{W_1} \times 100$$

Kinetics & ICH Testing

Accelerated testing (40°C / 75% RH) modeling reaction rates via Arrhenius equation.

$$k = A e^{-\frac{E_a}{RT}}$$

Tabeting Defects & Mitigations

Compression Mechanical Defects

- **Capping & Lamination:** Partial or complete air entrapment splitting. *Fix: Increase dwell time & reduce pre-compression speed.*
- **Sticking & Picking:** Material adherence to punch faces. *Fix: Increase Mg Stearate or dry granules to lower LOD.*

Flow & Mass Variation Defects

- **Weight Variation:** Poor powder die filling due to segregation. *Fix: Optimize particle size distribution & add colloidal silica.*
- **Mottling:** Uneven color distribution across surface. *Fix: Ensure homogeneous dye dispersion in binder solution.*

Quality Control & Critical Attributes

99.5%

Target Batch Release Compliance Rate

Dissolution Efficiency: Q > 80% at 30 minutes in simulated gastric fluid (pH 1.2).

Content Uniformity: Acceptance Value (AV) < 15 as mandated by USP <905>.

Hardness & Friability: Crushing strength 80–120 N; friability strictly < 0.8%.

Key Takeaways & Conclusion

Rational Excipient Design

Matching API physical properties with complementary fillers and disintegrants ensures robust compressibility and rapid dissolution.

Optimized Process Control

High-shear wet granulation paired with fluid bed drying mitigates flow variability and defect risks during tablet compression.

Quality by Design (QbD)

Incorporating moisture barrier film coating and strict humidity controls guarantees a stable 24+ month commercial shelf life.

Image Sources

http://www.skpharmteco.com/wp-content/uploads/2024/10/Dissolution-testing-equipment-and-testing-scaled-e1701115219153.jpg

Source: www.skpharmteco.com

https://www.fluidpack.net/admin/blog-images/high-speed-double-rotary-tablet-press-act-v.jpg

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